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Nitrogen and Phosphorus Budgets in Experimental Grasslands of Variable Diversity

Yvonne Oelmann,* Yvonne Kreutziger, Vicky M. Temperton, Nina Buchmann, Christiane Roscher, Jens Schumacher, Ernst-Detlef Schulze, Wolfgang W. Weisser, and Wolfgang Wilcke

ABSTRACT

Previous research has shown that plant diversity influences N and P cycles. However, the effect of plant diversity on complete ecosystem N and P budgets has not yet been assessed. For 20 plots of artificially established grassland mixtures differing in plant diversity, we determined N and P inputs by bulk and dry deposition and N and P losses by mowing (and subsequent removal of the biomass) and leaching from April 2003 to March 2004. Total deposition of N and P was 2.3 ± 0.1 and 0.2 ± 0.01 g m^{-2} yr^{-1}, respectively. Mowing was the main N and P loss. The net N and P budgets were negative (–6.3 ± 1.1 g N and –1.9 ± 0.2 g P m^{-2} yr^{-1}). For N, this included a conservative estimate of atmospheric N_2 fixation. Nitrogen losses as N_2O were expected to be small at our study site (<0.05 g m^{-2} yr^{-1}). Legumes increased the removal of N with the harvest and decreased leaching of NH_4–N and dissolved organic nitrogen (DON) from the canopy. Reduced roughness of grass-containing mixtures decreased dry deposition of N and P. Total dissolved P and NO_3–N leaching from the canopy increased in the presence of grasses attributable to the decreased N and P demand of grass-containing mixtures. Species richness did not have an effect on any of the studied fluxes. Our results demonstrate that the N and P fluxes in managed grassland are modified by the presence or absence of particular functional plant groups and are mainly driven by the management.

Atmospheric input of N and the resulting consequences for ecosystem stability in forests and grasslands have been extensively discussed during the recent decades (Lindberg et al., 1986; Schulze et al., 1989; Rihm and Kurz, 2001). Nitrogen inputs result in decreasing species richness and significant changes in the N cycle (Baron et al., 2000; Phoenix et al., 2003; Zak et al., 2004). Nitrogen deposition might shift ecosystems from N to P limitation. However, P deposition might also feed back on P cycling of ecosystems. Increased P availability might cause a loss of endangered plant species since these persist better at P-limited conditions (Wassen et al., 2005). Therefore, when addressing nutrient deposition in managed, species-rich grassland ecosystems, P might be similarly or even more important than N.

Inputs of N and P into a grassland system can be partitioned into wet and dry deposition. The roughness and the surface of the vegetation canopy, and the weather conditions, e.g., wind speed and temperature, control the extent of wet and dry deposition (Ulrich, 1983). Dry deposition is a relevant flux for net ecosystem budgets but cannot be directly determined because it is not possible to mimic the particle-capturing properties of a vegetation canopy. Therefore, indirect approaches have to be used. Usually, "bulk deposition" including wet deposition and coarse particulate deposition (which is gravitationally settling) is measured with Hellmann-type rain collectors. The fine particulate dry deposition on the surface of the vegetation canopy can be modeled with a micrometeorological approach (Lindberg et al., 1986; Hesterberg et al., 1996). Alternatively, a simple estimation method using measurements of bulk deposition, throughfall, and stemflow, and assuming as suggested by Ulrich (1983) that Cl^- is an inert tracer for forests and could be applied to grasslands, where stemflow, however, is difficult to determine and therefore neglected. If it is assumed that there is no N leaching in N-limited ecosystems, the part of the N flux in throughfall that cannot be explained by fine particulate and bulk deposition is roughly considered as gaseous N deposition (Ulrich, 1983). If N is taken up by the canopy, which can occur in N-limited systems, uptake is underestimated by the gaseous deposition. As gaseous N deposition is of minor importance for net N budgets, neglecting these fluxes only produces a small error of the total input estimate (Rihm and Kurz, 2001). Thus, the measurement of throughfall is a prerequisite to determine fine particulate dry deposition with the Ulrich model although throughfall also contains an internal flux because of canopy leaching. In addition to wet and dry deposition, fixation of atmospheric N_2 either by free-living bacteria or by symbiosis of legumes and bacteria is an important source of N in managed grassland ecosystems requiring large amounts of P (Burke et al., 1998; Jacot et al., 2000a, 2000b; Spehn et al., 2002).

Nitrogen and P are lost from the grassland system by leaching from the soil to the groundwater (Tilman et al., 1996; Scherer-Lorenzen et al., 2003; Toor et al., 2003).

Y. Oelmann, Inst. of Ecology, Dep. of Soil Science, Berlin Univ. of Technology, Salzufer 11-12, D-10587 Berlin, Germany. Y. Oelmann and W. Wilcke, Geographic Inst., Professorship of Soil Geography/Soil Science, Johannes Gutenberg Univ., Johann-Joachim-Becherweg 21, D-55128 Mainz, Germany. Y. Kreutziger, Inst. of Geography, Friedrich Schiller Univ., Löbdergraben 32, D-07743 Jena, Germany. V.M. Temperton, Inst. of Chemistry and Dynamics of the Geosphere, ICGIII Phytosphere Inst., Jülich Research Centre, D-52425 Jülich, Germany. N. Buchmann, Inst. of Plant Sciences, ETH Zentrum LFW C56, Universitätsstrasse 2, CH-8092 Zurich, Switzerland. C. Roscher, J. Schumacher, and E.-D. Schultze, Max Planck Inst. for Biogeochemistry, P.O. Box 100164, D-07701 Jena, Germany. W.W. Weisser, Inst. of Ecology, Friedrich Schiller Univ. of Jena, Dornburger Straße 159, D-07743 Jena, Germany. Received 2 June 2006. *Corresponding author (yvonne.oelmann@uni-mainz.de).

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Abbreviations: ANOVA, analysis of variance; cLEA, leaching from the canopy; DD, dry deposition; DON, dissolved organic N; DOP, dissolved organic P; DRP, dissolved reactive P; FDR, frequency domain reflectometry; GLM, general linear model; sLEA, leaching from soil; TD, total deposition; TDN, total dissolved N; TDP, total dissolved P; TFD, throughfall deposition; VWM, volume-weighted mean; BD, bulk deposition.

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Downward water fluxes are mainly driven by the volume of precipitation, temperature, evapotranspiration, wind speed, and soil water storage. For N, denitrification particularly during anaerobic conditions in soil and freeze-thaw intervals leads to gaseous N losses as N_2, N_2O, or NO_x (Kammann et al., 1998; Glatzel and Stahr, 2001; Flechard et al., 2005). Depending on the productivity of the managed grassland, high amounts of N and P can be removed from the system by mowing and subsequent removal of the biomass (Olde Venterink et al., 2002).

In ecosystem budget estimates, all fluxes are quantified and the difference between the sum of inputs and that of outputs is termed net budget. Figure 1 illustrates the fluxes considered in our study. Losses or gains give a first hint on how nutrient availability in the ecosystem will develop in the future (Phoenix et al., 2003). The knowledge of the contribution of the deposition fluxes to the net budget can help in assessing the effect of atmospheric deposition on ecosystem stability. We hypothesize that bulk and dry deposition contribute substantially to net N and P budgets.

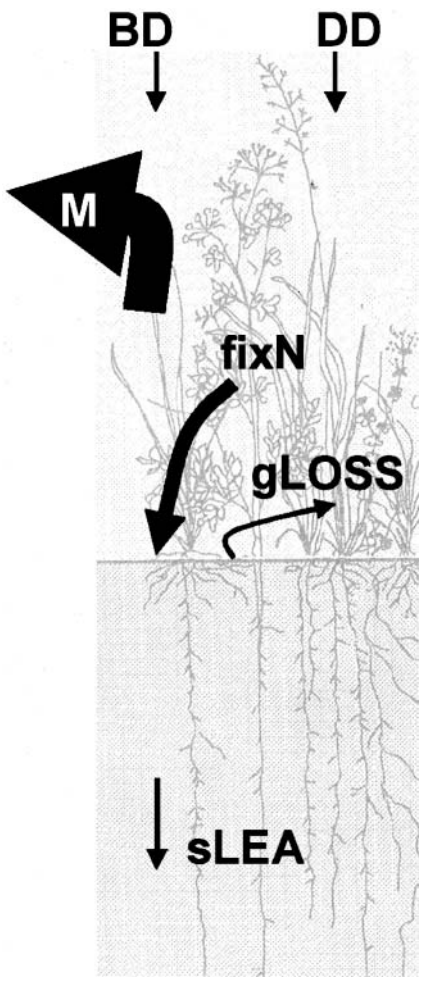

Fig. 1. Conceptual diagram of the in- and outputs in the studied grassland. BD = bulk deposition (wet and coarse particulate); DD = dry deposition (fine particulate); M = mowing; sLEA = leaching from soil; fixN = fixation of atmospheric N_2; gLOSS = gaseous N loss.

The globally increasing extinction of species has raised the question whether particular functional plant groups or plant diversity per se control nutrient cycling in ecosystems (Schulze and Mooney, 1993; Chapin et al., 2000; Loreau et al., 2001). The underlying hypothesis states that the presence of different functional groups and increasing diversity result in complementary and more efficient resource use and correspondingly in tighter element cycles (Tilman et al., 1996). Net N and P budgets in managed grasslands may be influenced by particular functional plant groups because of their specific traits. As an example, legumes often have a strong impact on productivity and plant and soil N (Ledgard and Steele, 1992; Mulder et al., 2002; Spehn et al., 2002) because they provide an additional N input to the grassland. In contrast to legumes, most of the grass species might have a negative effect on N pools in above ground biomass, and on plant-available N in soil due to their low N concentrations in aboveground biomass and their extensive rooting system (Hooper, 1998; Craine et al., 2002). Recent studies have shown that a high plant diversity per se resulted in increasing plant N pools in aboveground biomass and this was related to decreasing N leaching from soil (Scherer-Lorenzen et al., 2003). In the only study we know on the influence of diversity on P cycles, Hooper and Vitousek (1998) found no effect of plant diversity on P availability in soil, but they did not address P fluxes. Up to now, all published studies relating plant diversity and nutrient cycling focused on particular compartments of N and P cycling in managed grassland systems. However, to test the hypothesis of tighter nutrient cycles at higher diversity in differently diverse grassland systems, a complete N and P balance has to be considered (Best and Jacobs, 2001; Olde Venterink et al., 2002; Phoenix et al., 2003). We hypothesize that net N and P budgets are influenced by the presence of legumes or grasses and indicate tighter nutrient cycles with increasing diversity.

The objectives of this study were: (i) to estimate N and P net budgets based on the N and P input by bulk and dry deposition (for N additionally by N fixation) and element output by mowing and subsequent removal of biomass and leaching from soil in experimental grassland mixtures of variable diversity, and (ii) to assess the effect of functional plant groups and species diversity on the N and P fluxes and net budgets in these grassland mixtures.

METHODS

This study is part of a large-scale grassland experiment (known as "The Jena Experiment") which addresses the role of biodiversity in element cycling and trophic interactions in a grassland ecosystem (Roscher et al., 2004).

Study Site

The field site is located near the German city of Jena (50°55′ N; 11°35′ E; 130 m above sea level). Mean annual air temperature is 9.3°C, and mean annual precipitation is 587 mm (Kluge and Müller-Westermeier, 2000). The soil is a Eutric Fluvisol developed from up to 2 m-thick fluvial sediments which are almost free of stones. Due to fluvial dynamics, the

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texture ranges from sandy loam near the river to silty clay with increasing distance from the river. This systematic variation in soil texture is considered in the statistical design of the experiment by arranging the plots in four blocks parallel to the river on texturally homogeneous subareas and including the block effect into the statistical analyses. Down to 0.3-m soil depth, organic C concentrations range from 13 to 33 g kg^{-1}, organic C/N ratios from 8 to 15, and pH (H_2O) from 7.1 to 8.4. The soil contains some carbonates (15 g kg^{-1} CO_3^{2-}–C, 0- to 0.3-m soil depth). The site was used as an arable field for the last 40 yr before the experiment. It was converted from grassland in the early 1960s, regularly plowed, probably occassionally irrigated, and fertilized over the last decades for the growing of vegetables and wheat. However, in autumn (mean of sampling campaigns in October 2002, 2003, and 2004) plant-available mineral N (KCl extractable $NO_3^- + NH_4^+$) amounted to 0.9 g N m^{-2} indicating that the influence of former fertilization can be neglected in 2003 and 2004.

The complete experimental design is described in Roscher et al. (2004). Briefly, the main experiment is comprised of 86 plots (20 by 20 m) of different levels of species richness (0, 1, 2, 4, 8, 16, 60) and different numbers (0, 1, 2, 3, 4) of functional groups (grasses, small herbs, tall herbs, legumes) chosen by the random replacement method from a species pool of 60 species from the Molinio-Arrhenatheretea meadows, Arrhenatherion community (Ellenberg, 1996). In this study, we refer to the Block 2 (20 plots) including a subset of replicates representing the complete range of diversity levels (1: $n = 4$, 2: $n = 4$, 4: $n = 4$, 16: $n = 3$, 60: $n = 1$; App. 1 to 3). The management of all plots was adapted to extensive meadows used for hay production and mown mechanically twice a year in June and September (after harvesting by scissors for the purpose of biomass production estimates). Plots were not fertilized during the experimental period. To maintain the sown species diversity level, plots were regularly weeded.

Sampling

In the second year after establishment of the experiment, we collected rainfall, throughfall, and cumulatively extracted soil solution every 2 wk from April 2003 to March 2004.

Rainfall and throughfall were collected with rain collectors (2-L sampling bottles connected to funnels [∅0.12 m], both polyethylene). The sampling bottles were protected against larger particles and small animals with a polyethylene net (0.005 m mesh width). The collectors were cleaned with deionized water before installation and replaced by clean collectors in 2- to 3-mo intervals. Rainfall collectors were installed 1 m above surface at two edges and in the middle of the field site, whereas each studied plot on Block 2 was equipped with throughfall collectors (in triplicates, $n = 60$). The throughfall sampling bottle was lowered into the soil, so that the uppermost edge of the funnel was at a height of 0.15 m above soil surface.

Five months before the collection period, suction plates (UMS, Munich, Germany, sintered glass, pore size 1 to 1.6 μm) were installed at a depth of 0.3 m ($n = 60$) and coupled with a permanent vacuum system to collect soil solution. The vacuum system was regulated with the help of manual measurements of soil matric potential. In summer 2003, we were unable to extract soil solution because of dry soil conditions. Soil water content was measured at a depth of 0.35 m using the FDR profile probe PR1/6 (Delta-T, Cambridge, UK) once per week. To maximize the accuracy of the measurements the probes were calibrated for each site with the help of field measurements of matric potential and water retention curves determined in the laboratory.

Above ground plant biomass was harvested by hand on all plots within a frame (0.2 by 0.5 m, height 0.03 m) with four randomly located replicates per plot in May and August 2003 before the mechanical mowing with subsequent complete removal of the mown biomass.

Chemical Analyses

Rainfall and throughfall samples were filtered through ashless filters (pore size 4 to 7 μm, No. 790, Schleicher and Schuell, Dassel, Germany) and afterward all samples were kept frozen until analysis. We did not add poison to the collectors, but we never observed any growth of microorganisms in our solutions. In rainfall, throughfall, and soil solution we determined the concentrations of NO_3^-, NH_4^+, and total dissolved N (TDN), and dissolved reactive P (DRP) and total dissolved phosphorus (TDP). In rainfall and throughfall we additionally measured chloride (Cl^-) concentrations.

Nitrate and NH_4^+ concentrations were measured in the respective solutions with a continuous flow analyzer (CFA, Skalar, Breda, The Netherlands). Nitrate was analyzed photometrically after reduction to NO_2^- and reaction with sulfanilamide and naphthylethylenediamine-dihydrochloride to an azo-dye. Our NO_3^- concentrations contained an unknown contribution of NO_2^- that is expected to be small. Simultaneously to the NO_3^- analysis, NH_4^+ was determined photometrically as 5-aminosalicylate after a modified Berthelot reaction. The detection limits of NO_3^- and NH_4^+ were 0.02 and 0.03 mg N L^{-1}, respectively. Total dissolved N in soil solution was analyzed by oxidation with $K_2S_2O_8$ followed by reduction to NO_2^- as described above for NO_3^-. Dissolved organic N (DON) concentrations in soil solution were calculated as the difference between TDN and the sum of mineral N ($NO_3^- + NH_4^+$). In 5% of the samples, TDN was equal to or smaller than mineral N. In these cases, DON was assumed to be zero. Dissolved reactive P in the soil solution was measured photometrically with a CFA. Ammonium molybdate catalyzed by antimony tartrate reacts in an acidic medium with phosphate and forms a phospho-molybdic acid complex. Ascorbic acid reduces this complex to an intensely blue-colored complex. Total dissolved P in soil solution was analyzed by irradiation with UV and oxidation with $K_2S_2O_8$ followed by reaction with ammonium molybdate described above for DRP. As the molybdic complex forms under strongly acidic conditions, we could not exclude the hydrolysis of labile organic P compounds in our samples. Furthermore, the molybdate reaction is not sensitive for condensed phosphates. The detection limits of both TDP and DRP were 0.02 mg P L^{-1}, respectively. Dissolved organic P (DOP) in soil solution was calculated as the difference between TDP and DRP. In 17% of the samples, TDP was equal to or smaller than DRP; in these cases, DOP was assumed to be zero. To estimate fine particulate dry deposition chloride concentrations in rainfall and throughfall were determined with a capillary electrophoresis (G 1600, Hewlett-Packard, Waldbronn, Germany). The detection limit of Cl^- measurements was 0.1 mg Cl L^{-1}.

Community plant biomass from all investigated plots ($n = 20$) was harvested and separated into species. After oven-drying (70°C, 48 h) to constant weight, plant material was weighed giving a dry weight for all species in all plots. To determine N in aboveground plant communities, living plant material of all subsamples per plot and per harvest campaign were ground with a Cyclotec 1093 Sample Mill (Foss Tecator, Hoganas, Sweden) using a 0.5-mm screen for chemical analysis. Twenty mg of the ground plant material was analyzed for plant N concentration with an elemental analyzer CE 1110 (Carlo Erba Instruments, Milan, Italy). Plant community N pools were then

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calculated using community biomass and N concentrations. To determine P concentrations in above ground biomass, 100 mg of each sample was digested with 5 mL of HNO_3 at 260°C and at about 60 to 70 bar using the multiwave-assisted high pressure digestion unit "Multiwave" (PerkinElmer, Rodgau-Jügesheim, Germany). Concentrations of P in the extract were measured with inductively coupled plasma-atomic emission spectrometry (ICP–AES) (Optima 3300 DV, PerkinElmer, Rodgau-Jügesheim, Germany). We did not assess P concentrations in all samples (32% of the studied plots). However, there was a strong correlation between harvested biomass and P concentrations ($y = 0.0034$ biomass (g m^{-2} yr^{-1}); $r = 0.96$, $p < 0.01$). Therefore, we used this equation to calculate P pools in plots where P was not measured.

Calculations and Statistical Analyses

We calculated volume-weighted mean (VWM), TDN, and TDP concentrations for each plot for the study period (April 2003 to March 2004) using the weight of collected volumes of soil solution.

To estimate the dry deposition and "canopy" leaching the model of Ulrich (1983) was used. In this model Cl^- is considered as an inert tracer for fine particulate dry deposition (i.e., it is assumed that Cl^- is neither taken up nor leached from the plants). The total deposition (TD) of an element i was calculated with Eq. [1]:

$$TD_i = BD_i + DD_i \quad [1]$$

where BD refers to wet + coarse particulate deposition measured above the canopy and DD is fine particulate dry deposition estimated with Eq. [2]. The estimate of DD does not include gaseous deposition for N. However, the latter is included in our estimate of canopy leaching (see below).

$$DD_i = \left(\frac{TFD_{Cl}}{BD_{Cl}} BD_i\right) - BD_i \quad [2]$$

In this equation TFD_{Cl} represents the throughfall deposition of Cl^-. According to Ulrich (1983), the sum of throughfall and stemflow deposition of Cl^- is related to the throughfall deposition of Cl^-. However, because of technical difficulties in measuring stemflow of a grassland ecosystem, we excluded stemflow from the model. The quotient of TFD_{Cl}/BD_{Cl} is called the deposition ratio.

As internal fluxes might be influenced by particular functional plant groups, we also calculated "canopy" leaching (cLEA) based on Eq. [3] (again excluding stemflow deposition).

$$cLEA_i = TFD_i - BD_i - DD_i \quad [3]$$

These leaching rates include an unknown contribution of dry gaseous deposition of N. If N is taken up by the canopy (i.e., cLEA assumes negative values), the uptake is underestimated by the gaseous deposition.

To determine element leaching to groundwater from soil layers deeper than 0.35-m soil depth, we estimated water fluxes based on a simple model (Kreutziger, 2006). We assume that the main rooting zone is established between the 0- and 0.35-m soil depths. At our study site, 80 to 90% of root biomass was observed in this depth (H. Beßler and C. Engels, personal communication, 2006), which is in line with values reported in the literature (Jackson et al., 1996). The amount of leached water (LW, mm) was computed using Eq. [4]:

$$LW = \Delta S + P - AET \quad [4]$$

where ΔS is the difference in water storage in soil between two subsequent sampling dates (t1 − t2), P is rainfall, and AET is actual evapotranspiration. Total water stored (S) in the soil column up to 0.35 m was calculated for each site and measurement date using the volumetric water content readings of the FDR pipe. Daily potential evapotranspiration (PET) was estimated using the Penman-Wendling equation (DVWK, 1996) based on rainfall, air temperature, relative humidity, radiation, saturated/actual water vapor pressure, and wind speed at 2 m height data of the central weather station on the field site. Actual evapotranspiration (AET) and fluxes were calculated according to Eq. [5] through [7]. Furthermore, we calculated capillary rise termed "upward flux" (UF) using Eq. [7]:

$$\text{if } \Delta S + P < PET \text{ then } AET = \Delta S + P \text{ and } LW = 0 \quad [5]$$

$$\text{if } \Delta S + P > PET \text{ then } AET = PET, \quad LW = \Delta S + P - PET \quad [6]$$

$$\text{if } \Delta S < 0 \text{ and } |\Delta S| > P \text{ then } UF = \Delta S - P \text{ with } AET = 0 \text{ and } LW = 0 \quad [7]$$

The amount of yearly leached water was calculated as the sum of the weekly calculations for the whole study period. The computed fluxes are estimates because of the low frequency of water content measurements. Leached element mass was calculated by multiplying volume-weighted element concentrations and calculated water fluxes during the study period.

The net budgets (nB) of N and P were calculated as the difference between the sum of inputs and the sum of outputs (Fig. 1; Eq. [8]).

$$nB = TD\ (+\ fixN) - M - sLEA \quad [8]$$

where TD is total deposition, M export with mowing, sLEA leaching from soil, and fixN N fixation by symbiosis between legumes and bacteria. The N fixation is roughly estimated by assuming that legumes contribute as much N to the total aboveground N pool as they contribute biomass. This might be an underestimation because legumes have higher N concentrations than other functional groups, but not all N in legumes has been fixed from the atmosphere. However, no direct measures of the contribution of legumes to the aboveground N pool were available. Measurements of gaseous N losses by denitrification are not explicitly included in our investigation except for two local short-term measurements of N_2O. Closed static chambers were installed in June 2003. Chamber air (30 mL) was sampled by syringes into pre-evacuated 20-mL vials in July and August 2003. Samples were withdrawn at regular intervals during chamber closure. Chambers were closed for 40 min. (N_2O was determined by gas chromatography Agilent 6890 with ECD detector [Agilent, Waldbronn, Germany]).

Before statistical analyses, variance homogeneity of the data sets was tested with the Levene-Test (SPSS, 2003). In case of heteroscedasticity we used the Welch test. To test for the effect of particular functional plant groups, we used a t test. Additionally, we conducted a univariate ANOVA including biomass as a covariable before fitting the respective functional groups as a fixed factor (general linear model [GLM], type I sum of squares). Similarly, a univariate ANOVA was performed on the respective data sets using species richness as a fixed factor.

RESULTS

Net Budget

During the study period the sum of cumulatively collected rainfall and throughfall was 487 and 492 mm,

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Table 1. Volumes and volume-weighted mean (VWM) total N, total P, and Cl^- concentrations in the respective solutions averaged across all investigated plots. Values are given as mean ± SE.

Parameter	Volume	VWM N	VWM P	VWM Cl^-
	mm yr^{-1}	mg L^{-1}		
Rainfall	487 ± 1.5	2.7 ± 0.3	0.3 ± 0.1	1.2 ± 0.2
Throughfall	492 ± 4.7	3.7 ± 0.4	0.4 ± 0.1	1.6 ± 0.1
Soil solution	107 ± 2.2	1.0 ± 0.1	0.1 ± 0.0	NA†

† NA = not available.

respectively (Table 1). We did not observe any interception loss. Ranges of VWM concentrations of N species in rainfall were: 0.9 to 1.1 mg NO_3–N L^{-1}, 1.3 to 1.7 mg NH_4–N L^{-1}, and 0.3 to 0.6 mg DON L^{-1}. The mean contribution of the respective N species to TDN was 38% (NO_3–N), 46% (NH_4–N), and 16% (DON). We observed a range of 0.1 to 0.4 mg L^{-1} for VWM DRP in rainfall resulting in a mean contribution of 90% to VWM TDP. Volume-weighted mean DOP concentrations were negligible (0.02 to 0.03 mg L^{-1}). Based on the relation of VWM Cl^- concentrations in rainfall and in throughfall of each plot we found annual deposition ratios ranging between 1.2 and 2.4. Therefore, the maximum fine particulate dry deposition estimate was 2.4 times bulk deposition. Estimated fine particulate dry deposition of N and P was 1.0 g N m^{-2} and 0.1 g P m^{-2}, respectively (Table 2).

In our study, legumes contributed 19 ± 7% to biomass production. The calculated fixation of atmospheric N_2 ranged from to 0 to 28.8 g N m^{-2} yr^{-1} (mean across all plots ± SE: 4.2 ± 1.8 g N m^{-2} yr^{-1}).

Averaged across all plots, 612 ± 73 g m^{-2} plant biomass were removed by mowing in 2003. Mean concentrations in biomass were for N: 19 ± 1 mg g^{-1} (May 2003) and 23 ± 1 mg g^{-1} (August 2003), and for P: 4.0 ± 0.4 mg g^{-1} (May 2003) and 3.5 ± 0.5 mg g^{-1} (August 2003). The average yearly removal by mowing was 12.7 ± 1.9 g N m^{-2} and 2.1 ± 0.2 g P m^{-2}. The calculation of the removal is based on the biomass produced in 2003 resulting in a close correlation between mown and removed biomass and N and P removal (N: $r = 0.96$, $p < 0.001$; P: $r = 0.99$, $p < 0.001$).

Table 3. Nitrate nitrogen, NH_4–N, and dissolved organic N (DON) deposition and leaching. Values are given as mean ± SE. Legumes (Le) or grasses (Gr) following the values indicate significant differences between presence/absence of the respective functional plant group (*t* test, $p < 0.05$).

Flux†	NO_3–N	NH_4–N	DON
	g m^{-2} yr^{-1}		
TD	0.9 ± 0.04	1.1 ± 0.1	0.4 ± 0.02
BD	0.5 ± 0.1	0.6 ± 0.1	0.2 ± 0.04
DD	0.4 ± 0.04Gr	0.5 ± 0.1Gr	0.2 ± 0.02Gr
cLEA‡	−0.4 ± 0.1Gr	−0.2 ± 0.2Le	0.04 ± 0.04Le
TFD	0.5 ± 0.03Gr	1.0 ± 0.2Le	0.4 ± 0.04Le
sLEA	0.13 ± 0.07	0.00 ± 0.00	0.09 ± 0.01

† TD = total deposition; BD = bulk deposition; DD = dry deposition; cLEA = leaching from the canopy; TFD = throughfall deposition; sLEA = leaching from soil.
‡ Positive values refer to leaching from the canopy, whereas negative values indicate uptake by the canopy.

Water leached from soil accounted for 22% of water input by rainfall (82.6 to 118.8 mm). Volume-weighted mean TDN and TDP concentrations in soil solution ranged from 0.7 to 4.5 mg N L^{-1} and not detected to 0.1 mg P L^{-1} (Table 1). During the study period 0.1 to 0.4 g N m^{-2} and 0 to 0.01 g P m^{-2} had leached from the soil (Table 2). Nitrogen mainly leached from soil as DON (Table 3), NO_3–N contributed 30% to total N leaching, whereas NH_4–N did not contribute to total N leaching (Table 3).

Calculated net budgets of N and P in the investigated grassland plots were in a range of –13.5 to +0.8 g N m^{-2} (mean ± SE: –6.3 ± 1.1 g N m^{-2} yr^{-1}) and –3.4 to 0.0 g P m^{-2} (Fig. 2). Nitrogen leaching from soil contributed 0.4 to 8.9%, whereas P leaching contributed 0.1 to 1.7% to the total export of N and P (sum of mowing and leaching). Input via atmospheric deposition reduced the negative N budget by 17.9% and for P by 11.6%. Increasing biomass resulted in decreasing N and P net budgets (= increasing loss of N and P; Fig. 3).

Table 2. Nitrogen and P deposition and leaching classified according to the presence (+) or absence (–) of legumes (+: $n = 13$; –: $n = 7$) or grasses (+: $n = 8$; –: $n = 12$). Values are given as mean ± SE. Different letters following the values indicate significant differences between presence/absence of the respective functional plant group (*t* test, $p < 0.05$).

Flux†	Mean	+ Legumes	− Legumes	+ Grasses	− Grasses
	g m^{-2} yr^{-1}				
Nitrogen					
M	11.4 ± 1.8	15.3 ± 2.6a	8.3 ± 1.2b	15.8 ± 1.8	10.5 ± 2.7
TD	2.3 ± 0.1	2.3 ± 0.3	2.2 ± 0.4	2.0 ± 0.1a	2.5 ± 0.1b
BD	1.3 ± 0.1				
DD	1.0 ± 0.1	1.0 ± 0.1	0.9 ± 0.0	0.7 ± 0.1a	1.2 ± 0.1b
cLEA‡	−0.5 ± 0.2	−1.0 ± 0.4a	0.2 ± 0.3b	−0.1 ± 0.3	−0.7 ± 0.2
TFD	1.8 ± 0.2	1.4 ± 0.2a	2.5 ± 0.3b	1.9 ± 0.2	1.7 ± 0.2
sLEA	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0	0.1 ± 0.0	0.2 ± 0.0
Phosphorus					
M	2.2 ± 0.3	2.3 ± 0.3	1.7 ± 0.3	2.7 ± 0.2a	1.7 ± 0.3b
TD	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0	0.3 ± 0.0	0.2 ± 0.0
BD	0.1 ± 0.0				
DD	0.1 ± 0.0	0.1 ± 0.0	0.1 ± 0.0	0.1 ± 0.0a	0.1 ± 0.0b
cLEA‡	−0.1 ± 0.0	−0.1 ± 0.0	0.0 ± 0.0	0.0 ± 0.0a	−0.1 ± 0.0b
TFD	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0
sLEA	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00

† M = removal by mowing; TD = total deposition; BD = bulk deposition; DD = dry deposition; cLEA = leaching from the canopy; TFD = throughfall deposition; sLEA = leaching from soil.
‡ Positive values refer to leaching from the canopy, whereas negative values indicate uptake by the canopy.

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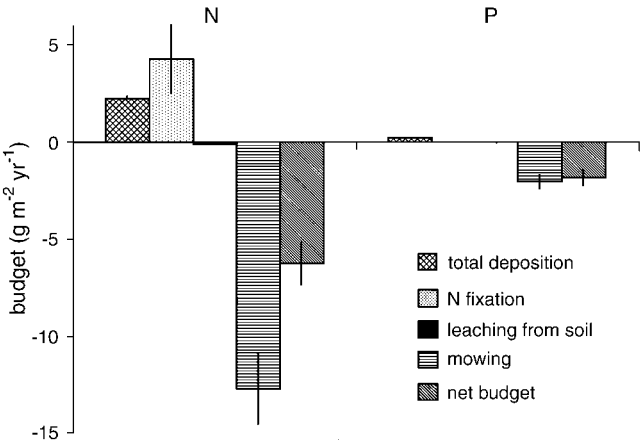

Fig. 2. Nitrogen and P balance including total deposition, N fixation (based on N yield of legumes), mowing, and leaching from soil. Gaseous N losses as NO_x or N_2O were supposed to be negligible.

Effect of Functional Groups and Species Richness

In May and August 2003, biomass N concentrations determined by harvesting representative subplots were higher in mixtures containing legumes than in mixtures without legumes ($t_{18} = 2.3$, $p = 0.04$). In mixtures containing legumes, less N was leached from the canopy (Table 2). This finding was mainly attributable to decreased leaching of NH_4–N when legumes were present (Table 3, $t_{7.3} = -2.5$, $p = 0.04$). To explore the effect of legumes without the possible interference with an effect of biomass we fitted annually mown biomass as a covariable before legume presence/absence in a univariate ANOVA, Type I. We observed a nonsignificant effect of mown biomass ($p > 0.05$) and a significant remaining effect of legumes (cLEA TDN: $F_1 = 13.5$, $p = 0.005$; cLEA NH_4–N: $F_1 = 10.7$, $p = 0.005$). The removal of N by mowing increased when legumes were present (Table 2) mainly because of increased mown biomass (ANOVA, Type I, $F_1 = 470.3$, $p < 0.000$) and increased N concentrations in mown biomass (fitted after annually mown biomass in an ANOVA Type I, $F_1 = 27.5$, $p < 0.000$). We did not find an effect of legume presence on P cycling ($p > 0.05$).

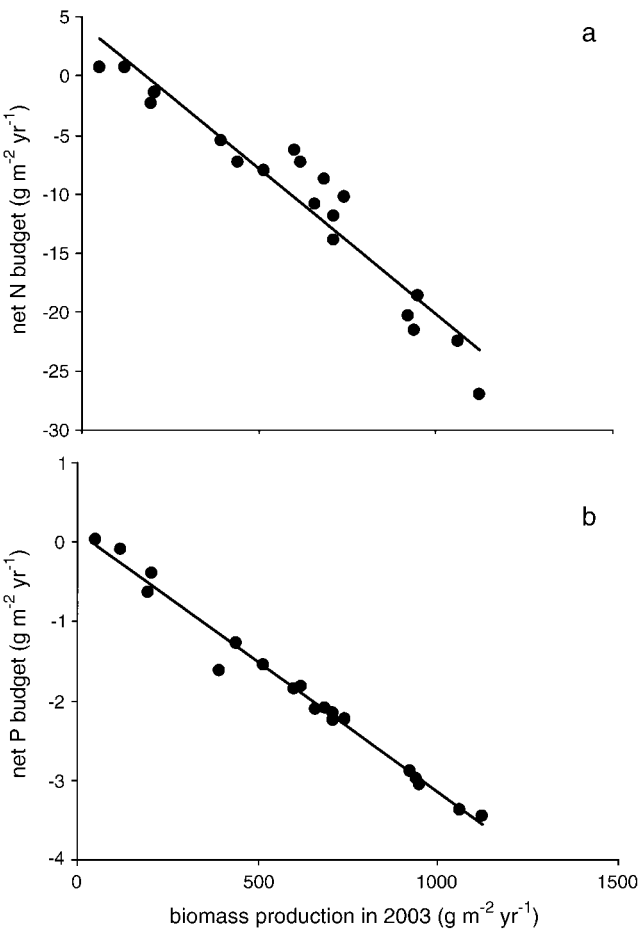

Fig. 3. Relationship between total biomass produced in 2003 and net N and P budgets. N: $r = 0.83$, $p < 0.001$; P: $r = 0.96$; $p < 0.001$.

The mown biomass in 2003 increased when grasses were present in a mixture ($t_{18} = 2.3$, $p = 0.03$). The presence of grasses decreased the deposition ratio, and therefore dry deposition of N and P (Table 2). The effect of grasses on the deposition ratio remained significant if annually mown biomass was fitted first in an ANOVA, Type I. If grasses were present in a mixture, on average less P was taken up by the canopy compared with mixtures without grasses (Table 2). Similarly, we observed decreased uptake of NO_3–N by the canopy in mixtures with grasses (Table 3, $t_{18} = 4.2$, $p = 0.001$). Decreased uptake was significantly related to the mown biomass of the grasses (ANOVA; TDP: $F_1 = 8.8$, $p = 0.009$; NO_3–N: $F_1 = 6.1$, $p = 0.025$). The presence of grasses increased the removal of P by mowing and decreased the P net budgets (= increased P loss) of the investigated mixtures (Table 2).

We did not find any significant effect of plant diversity on mown biomass, N and P concentrations in mown biomass, volume of throughfall, VWM concentrations in throughfall, dry deposition, throughfall deposition, leaching from the canopy, volume of water leached from soil, VWM concentrations in soil solution, leaching from soil, and net budgets in the studied plots (GLM; $p > 0.05$, data not shown).

Mown biomass (frequently used as a proxy for diversity in similar experiments) had an effect on internal fluxes. Volume-weighted mean TDP concentrations in throughfall correlated significantly positively with mown biomass in 2003 ($r = 0.59$, $p < 0.01$) and with TFD of P (Fig. 4a). Canopy leaching of P increased with increasing mown biomass in 2003 (Fig. 4b).

DISCUSSION

Net Budget

In 2003, mean annual rainfall was lower compared with the mean of rainfall between 1961 and 1990 of 587 mm (Müller-Westermeier and Riecke, 2004). The observed rainfall volumes are in line with results reported by the Thuringian Environmental and Geological Survey (TLUG, 2004). We did not detect any interception loss (Table 1) which is probably included in the error of sampling (rainfall standard deviation [SD] = 0.6%; throughfall SD = 4.3%). The high SD of throughfall samplers might be related to samplers which were not

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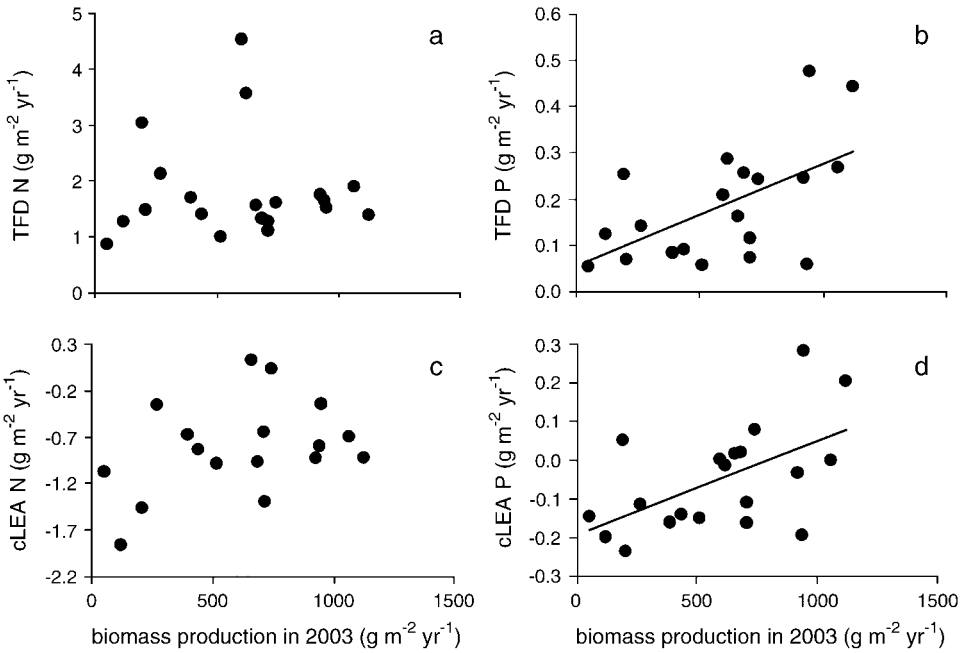

Fig. 4. Relationship between total biomass produced in 2003 and throughfall deposition (TFD) and leaching from the canopy (cLEA) of N (a, c) and P (b, d). TFD P: $r = 0.57$, $p < 0.01$; cLEA P: $r = 0.56$; $p < 0.01$.

completely covered by the canopy resulting in negligible interception losses. In a Brazilian cerrado, Lilienfein and Wilcke (2004) found that 0 to 9% of rainfall was evaporated from the canopy of grasslands during rainy and dry seasons. In the temperate zone, James et al. (2003) reported negligible interception losses particularly for grassland sites. Furthermore, possible interception losses in our study might partly be compensated by input through dew.

The calculated deposition ratio based on Cl^- deposition in throughfall and rainfall yielded a value of 1.8 ± 0.1 (average of three rainfall collectors) in our study. Using the same model for calculating dry deposition in the Brazilian savanna, Lilienfein and Wilcke (2004) found deposition ratios of 1.7 and 2.1 in a productive pure-grass and a degraded pasture system including some bushes, respectively. In the temperate zone (Europe), deposition ratios calculated based on micrometeorological modeling of dry deposition of N were 1.5 (winter crops; Goulding et al., 1998); 1.1 and 4.2 (grassland; Hesterberg et al., 1996; Rihm and Kurz, 2001). We are aware of possibly critical assumptions in the model we used: (i) Cl^- ions might be taken up by, or leached from the canopy (no inert tracer), and (ii) the deposition of other elements/compounds might depend on other factors than that controlling Cl^- deposition (transfer of Cl^- deposition ratio to other elements not possible). However, the application of the model of Ulrich (1983) in grassland ecosystems yielded deposition ratios comparable to micrometeorological estimates of N deposition. The model of Ulrich (1983) was originally developed for estimating element input in forested ecosystems. Our results show that this model offers an easily applicable method that can also be used for grasslands.

We found total (bulk + dry) N and P deposition of 2.3 and 0.2 g m^{-2} yr^{-1}, respectively (Table 2). The observed total N deposition is in the lower range of N deposition reported in other grassland studies (Best and Jacobs, 2001; Olde Venterink et al., 2002: 4.3 g m^{-2} yr^{-1}; Hesterberg et al., 1996: 1.8 g m^{-2} yr^{-1}; Rihm and Kurz, 2001: 1.1 g m^{-2} yr^{-1}). Due to technical difficulties we did not assess stemflow, nor did we find any estimate in the literature investigating stemflow in grasslands. Furthermore, we did not assess gaseous deposition of N as NH_3 or NO_x. In plots without legumes, we observed a small net average N leaching from the canopy of 0.2 g m^{-2}, which could be interpreted as gaseous N deposition if it is assumed that no N leaching from the canopy occurs in N-limited systems (Table 2). The small value of 0.2 g m^{-2} is a rough estimate for gaseous deposition, because the assumption that cLEA is zero might be incorrect. This finding suggests a low gaseous N deposition at our study site. If gaseous deposition is taken as 0.2 g m^{-2}, total N deposition at our study site would be 2.5 g N m^{-2}. In other grassland studies, gaseous deposition contributed 35 to 57% to total deposition in grasslands adjacent to fields in intensive agricultural use, where enhanced NH_3 emissions occurred (Hesterberg et al., 1996; Goulding et al., 1998; Rihm and Kurz, 2001). The adjacent meadows and pastures of our study site are also used for agriculture. However, due to the water reserve area restrictions, fertilizers have to be applied at low rates and manure application is forbidden (MLNU, 1997; TLVwA, 2004). Adding the high contributions of gaseous deposition to grassland systems reported in the literature to our total deposition results in 4.4 g N m^{-2} yr^{-1}, on average. The range of 2.5 to 4.4 g N m^{-2} covers the upper half of the range of N deposition reported in the literature (Best and Jacobs, 2001; Olde Venterink et al., 2002; Hesterberg et al., 1996; Rihm and Kurz, 2001). Phosphorus deposition from the atmosphere has rarely been investigated in grassland systems. Olde Venterink et al. (2002) reported total P deposition from

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the atmosphere of 0.02 g m^{-2} yr^{-1}, whereas Best and Jacobs (2001) found slightly higher total P deposition of 0.1 g m^{-2} yr^{-1}. Reviewing the literature Newman (1995) found total P deposition rates in various ecosystems across different climatic zones of 0.02 to 0.2 g m^{-2} yr^{-1}. Therefore, our total N and P deposition corresponds well with findings reported in the literature.

In our study, we used a conservative estimate for the input of N by atmospheric N_2–fixing legumes. Reviewing the literature, Carlsson and Huss-Danell (2003) reported that legumes (*Trifolium pratense* L., *T. repens* L., *Medicago sativa* L.) used for foraging derived 70 to 98% (average 80%) of the removal of N by mowing by fixation of atmospheric N_2. Jacot et al. (2000b) found fixation of atmospheric N_2 by legumes to contribute 59% to more than 90% to the removal of N by mowing in an Alpine grassland. They hypothesized that the observed decrease in the contribution of fixation to the removal of N by mowing with increasing altitude could mainly be explained by the proportion of legumes to total biomass (Jacot et al., 2000a). Provided that legumes completely relied on N_2 fixation to meet their N demand, and assuming that legumes contribute as much to the N removal by mowing as to the mown biomass, the removal of N by mowing of the legumes could be used as the maximum N fixation rate. If this were true, one might still underestimate N fixation because of (i) higher N concentrations in legume biomass compared with the mean N concentration of the plant mixture, (ii) facilitation, which will lead to removal of N derived from fixation with non-leguminous plant species, and (iii) loss of N originating from fixation by N-containing root exudates (McNeill and Wood, 1990; Whitehead, 1995). However, this error also applies to N fixation estimated with the help of $\delta^{15}N$ natural abundance or isotope dilution methods. Furthermore, the assumption of complete reliance of legumes on fixation overestimates N fixation, because legumes additionally take up N from soil. Therefore, we believe that the removal of N by mowing of legumes could serve as a conservative estimate of N fixation. Including maximum estimates of atmospheric N_2 fixation by legumes, 85% of all studied grassland mixtures lost N (Fig. 3), on balance, mainly because of mowing and subsequent removal of the mown biomass.

Among the studied in- and outputs, deposition from the atmosphere and leaching from soil did not contribute substantially to the N or P balance in our study (Fig. 2). Investigating highly productive wet meadows, Best and Jacobs (2001) and Olde Venterink et al. (2002) reported that deposition corresponded with 33 to 37% of N loss with the harvest, and with 1 to 7% of P loss. They also observed negligible N and P leaching from soil. Similarly, NO_3^- leaching from soil was low in calcareous and acidic grasslands (Di and Cameron, 2002; Phoenix et al., 2003; Scherer-Lorenzen et al., 2003).

Budgets for N and P at our study site were mainly governed by N and P removal with the mowing and were, therefore, closely negatively correlated with mown biomass (Fig. 3). Best and Jacobs (2001) and Olde Venterink et al. (2002) highlighted the importance of N and P removal by mowing in highly productive wet meadows where biomass production of 350 to 900 g m^{-2} yr^{-1} was comparable to our results.

We did not include N output as gaseous compounds like N_2, N_2O, or NO_x. These gases originate from nitrification and denitrification in soil depending on the availability of organic matter or NO_3^-, pH, soil moisture, and temperature (Granli and Bøckmann, 1994). Emissions of N_2O were measured twice at our sampling site (July and August 2003). Averaged across the two sampling dates, 6.2 ± 1.3 μg N m^{-2} h^{-1} were emitted as N_2O (A. Ekberg, personal communication, 2006). Assuming this emission to be constant throughout the day and the year and without accounting for spatial heterogeneity results in a very rough estimate of 0.05 g N m^{-2} yr^{-1}. Kammann et al. (1998), Glatzel and Stahr (2001), and Flechard et al. (2005) observed net N_2O fluxes of 0.02 g N m^{-2} yr^{-1} at extensively managed grasslands with comparable soil properties to our study site in terms of N availability and pH. Net NO_x fluxes are usually lower than net N_2O fluxes (Tilsner et al., 2003), and are therefore supposed to be negligible at our study site. Including the gaseous N_2O loss measured at our study site by Ekberg (personal communication, 2006) would result in a decrease of 0.5% of the net budget. However, N_2 might contribute substantially to total gaseous N losses (Bol et al., 2003; Mathieu et al., 2006), particularly under basic pH conditions at the study site (Simek et al., 2002).

Both mean N and P net budgets were negative indicating loss of N and P from the ecosystems. The N/P ratio of community biomass of all mixtures ranging from 3 to 14 in 2003 suggest that N limits plant growth (Koerselman and Meuleman, 1996). Given the annual net N losses, N could become more limiting in the long run.

Effect of Functional Groups and Species Richness

Legumes strongly influenced the N cycle, but no effect of legumes on the P cycle was detected. The additional N source of legumes (fixation of atmosperic N_2) resulted in higher N concentrations in legume-containing mixtures, thereby increasing N removal by mowing. We found decreased VWM TDN concentrations in throughfall and reduced N leaching from the canopy in mixtures containing legumes (Table 2), mainly because the canopy of legume-containing mixtures took up more NH_4–N than the canopy of mixtures without legumes (Table 3). We would have expected the opposite effect because increased N concentrations in the aboveground biomass in legume-containing mixtures might (i) be a result of the N fixation by legumes and might, therefore, reduce N uptake by the canopy, and (ii) favor herbivory resulting in increased leaf destruction and, therefore, in increased leaching of N from destroyed cells in the canopy. On the other hand, legume-containing mixtures showed an increased leaf area index (LAI) compared with plots without legumes (t test; e.g., sampling date: 22 Aug. 2003; vegetation height between 0.05 and 0.2 m: $t_{18} = 2.2$, $p = 0.04$; vegetation height between 0.2 and 0.35 m: $t_{18} = 1.7$, $p = 0.09$; measured with LAI-2000, LiCor, USA; data not shown). The increased surface area could lead to increased uptake of NH_4–N by the

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canopy in legume-containing mixtures. An alternative explanation is related to herbivory. Scherber et al. (2006) observed a positive effect of the presence of legumes on invertebrate herbivory at our study site. Investigating the effect of phytophagous insects on throughfall chemistry in forests, Stadler and Michalzik (2000) stressed the importance of immobilization of inorganic N species by insects. We speculate that decreased N leaching from the canopy is a result of the increased intake of NH_4–N by insects in mixtures containing legumes.

Grasses affected the N and P cycle. Dry deposition of P and N was lower in grass-containing mixtures (Table 2). The presence of grasses might decrease community roughness due to small and long leaves. We did not assess vegetation roughness in 2003. However, stratified height measurements across all investigated plots showed a decreased coefficient of variation in grass-containing mixtures in May 2005 ($t_{18} = -1.8$, $p = 0.08$; A. Weigelt and J. Schumacher, personal communication, 2006). This relative smoothness in grass-containing mixtures might cause decreased deposition of aerosols, and therefore, decreased dry deposition of N and P. Uptake of TDP and NO_3–N by the canopy decreased in the presence of grasses (Tables 2 and 3). Lower N and P demand of grasses compared with other functional groups and enhanced nutrient accessibility in soil due to the extensive rooting system in grass-containing mixtures might result in decreased uptake of NO_3–N and P by the canopy compared with mixtures without grasses (Marschner, 1995). Furthermore, the significant effect of biomass on leaching from the canopy might be explained by the increased biomass in grass-containing mixtures, increasing leaching from the canopy and resulting in decreased uptake by the canopy. Increased mown biomass resulted in increased removal of P by mowing in grass-containing mixtures. Therefore, more P was lost in grass-containing mixtures.

The investigated grassland mixtures did not show a correlation between species richness and mown biomass, maybe because of the small number of replicates. The effect of species richness on N pools in biomass and soil in similar experiments was mainly driven by increasing biomass with increasing diversity (Mulder et al., 2002; Spehn et al., 2002). As we did not find such a correlation, we neither observed a diversity effect on N or P concentrations in the respective solutions nor on N or P net budgets. However, for the whole Jena experiment with four blocks of which we studied only one, a significant diversity effect on biomass existed (Roscher et al., 2005). Thus, increased biomass production with increasing species richness would result in increasingly negative net budgets, because of the increased removal of nutrients by mowing. This would support the hypothesis of increasingly tighter nutrient cycling with increasing diversity in systems with less nutrient export by mowing.

CONCLUSIONS

(1) Bulk and dry deposition of N and P were of minor importance for the net budgets of the studied grassland mixtures. We observed that overall loss of N and P was mainly driven by mowing and subsequent removal of mown biomass. Particularly great N losses imply the increasing importance of N for plant growth in the studied grassland in the long-term perspective.

(2) Legumes had an effect only on the N cycle, whereas grasses influenced both N and P cycling. Species richness did not influence the studied parameters. However, taking all experimental plots (not only one of four blocks) into account, the existing positive correlation between plant species richness and aboveground biomass production might imply increased N and P losses in more diverse systems. The negative relationship between net budgets and biomass production indicate that in natural grasslands not subjected to mowing more effective resource use in diverse systems likely results in tighter element cycles.

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REFERENCES

Baron, J.S., H.M. Rueth, A.M. Wolfe, K.R. Nydick, E.J. Allstott, J.T. Minear, and B. Moraska. 2000. Ecosystem responses to nitrogen deposition in the Colorado Front Range. Ecosystems 3:352–368.

Best, E.P.H., and F.H.H. Jacobs. 2001. Production, nutrient availability, and elemental balances of two meadows affected by different fertilization and water table regimes in The Netherlands. Plant Ecol. 155:61–73.

Bol, R., S. Toyoda, S. Yamulki, J.M.B. Hawkins, and L.M. Cardenas. 2003. Dual isotope and isotopomer ratios of N_2O emitted from a temperate grassland soil after fertiliser application. Rapid Commun. Mass Spectrom. 17:2550–2556.

Burke, I.C., W.K. Lauenroth, M.A. Vinton, P.B. Hook, R.H. Kelly, H.E. Epstein, M.R. Aguiar, M.D. Robles, M.O. Aguilera, K.L. Murphy, and R.A. Gill. 1998. Plant-soil interactions in temperate grasslands. Biogeochemistry 42:121–143.

Carlsson, G., and K. Huss-Danell. 2003. Nitrogen fixation in perennial forage legumes in the field. Plant Soil 253:353–372.

Chapin, F.S., E.S. Zavaleta, V.T. Eviner, R.L. Naylor, P.M. Vitousek, H.L. Reynolds, D.U. Hooper, S. Lavorel, O.E. Sala, S.E. Hobbie, M.C. Mack, and S. Diaz. 2000. Consequences of changing biodiversity. Nature 405:234–242.

Craine, J.M., D. Tilman, D. Wedin, P. Reich, M. Tjoelker, and J. Knops. 2002. Functional traits, productivity and effects on nitrogen cycling of 33 grassland species. Funct. Ecol. 16:563–574.

Di, H.J., and K.C. Cameron. 2002. Nitrate leaching in temperate agroecosystems: Sources, factors, and mitigating strategies. Nutr. Cycl. Agroecosyst. 64:237–256.

DVWK. 1996. Ermittlung der Verdunstung von Land- und Wasserflächen. DVWK-Merkblätter zur Wasserwirtschaft 238.

Ellenberg, H. 1996. Die Vegetation Mitteleuropas mit den Alpen in ökologischer, dynamischer und historischer Sicht. 5th ed. Ulmer, Stuttgart.

Flechard, C.R., A. Neftel, M. Jocher, C. Ammann, and J. Fuhrer. 2005.

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Bi-directional soil/atmosphere N_2O exchange over two mown grassland systems with contrasting management practices. Global Change Biol. 11:2114–2127.

Glatzel, S., and K. Stahr. 2001. Methane and nitrous oxide exchange in differently fertilised grassland in southern Germany. Plant Soil 231:21–35.

Goulding, K.W.T., N.J. Bailey, N.J. Bradbury, P. Hargreaves, M. Howe, D.V. Murphy, P.R. Poulton, and T.W. Willison. 1998. Nitrogen deposition and its contribution to nitrogen cycling and associated soil processes. New Phytol. 139:49–58.

Granli, T., and O.C. Bøckmann. 1994. Nitrous oxide from agriculture. Norw. J. Agric. Sci. 12:1–128.

Hesterberg, R., A. Blatter, M. Fahrni, M. Rosset, A. Neftel, W. Eugster, and H. Wanner. 1996. Deposition of nitrogen-containing compounds to an extensively managed grassland in central Switzerland. Environ. Pollut. 91:21–34.

Hooper, D.U. 1998. The role of complementarity and competition in ecosystem responses to variation in plant diversity. Ecology 79:704–719.

Hooper, D.U., and P.M. Vitousek. 1998. Effects of plant composition and diversity on nutrient cycling. Ecol. Monogr. 68:121–149.

Jackson, R.B., J. Canadell, J.R. Ehleringer, H.A. Mooney, O.E. Sala, and E.-D. Schulze. 1996. A global analysis of root distributions for terrestrial biomes. Oecologia 108:389–411.

Jacot, K.A., A. Lüscher, J. Nösberger, and U.A. Hartwig. 2000a. The relative contribution of symbiotic N_2 fixation and other nitrogen sources to grassland ecosystems along an altitudinal gradient in the Alps. Plant Soil 225:201–211.

Jacot, K.A., A. Lüscher, J. Nösberger, and U.A. Hartwig. 2000b. Symbiotic N_2 fixation of various legume species along an altitudinal gradient in the Swiss Alps. Soil Biol. Biochem. 32:1043–1052.

James, S.E., M. Partel, S.D. Wilson, and D.A. Peltzer. 2003. Temporal heterogeneity of soil moisture in grassland and forest. J. Ecol. 91:234–239.

Kammann, C., L. Grunhage, C. Muller, S. Jacobi, and H.J. Jager. 1998. Seasonal variability and mitigation options for N_2O emissions from differently managed grasslands. Environ. Pollut. 102:179–186.

Kluge, G., and G. Müller-Westermeier. 2000. Das Klima ausgewählter Orte der Bundesrepublik Deutschland: Jena. Berichte des Deutschen Wetterdienstes 213.

Koerselman, W., and A.F.M. Meuleman. 1996. The vegetation N/P ratio: A new tool to detect the nature of nutrient limitation. J. Appl. Ecol. 33:1441–1450.

Kreutziger, Y. 2006. Rückkopplungseffekte verschieden diverser Grünlandökosysteme auf die Komponenten des Bodenwasserhaushaltes an einem Auestandort der Saale–Ergebnisse des Jenaer Biodiversitätsexperiments. Ph.D. thesis. Friedrich Schiller University, Jena, 174 S.

Ledgard, S.F., and K.W. Steele. 1992. Biological nitrogen fixation in mixed legume grass pastures. Plant Soil 141:137–153.

Lilienfein, J., and W. Wilcke. 2004. Water and element input into native, agri-, and silvicultural ecosystems of the Brazilian savanna. Biogeochemistry 68:131–133.

Lindberg, S.E., G.M. Lovett, D.R. Richter, and D.W. Johnson. 1986. Atmospheric deposition and canopy interaction of major ions in a forest. Science 231:141–145.

Loreau, M., S. Naeem, P. Inchausti, J. Bengtsson, J.P. Grime, A. Hector, D.U. Hooper, M.A. Huston, D. Raffaelli, B. Schmid, D. Tilman, and D.A. Wardle. 2001. Biodiversity and ecosystem functioning: Current knowledge and future challenges. Science 294:804–808.

Marschner, H. 1995. Mineral nutrition of higher plants. 2nd ed. Academic Press, London, San Diego, New York.

Mathieu, O., J. Leveque, C. Henault, M.J. Milloux, F. Bizouard, and F. Andreux. 2006. Emissions and spatial variability of N_2O, N_2, and nitrous oxide mole fraction at the field scale, revealed with N-15 isotopic techniques. Soil Biol. Biochem. 38:941–951.

McNeill, A.M., and M. Wood. 1990. ^{15}N estimates of nitrogen fixation by white clover (*Trifolium repens* L.) growing in a mixture with ryegrass (*Lolium perenne* L.). Plant Soil 128:265–273.

MLNU. 1997. ThürGewQualVO-Thüringer Verordnung über die Qualitätsanforderungen an Oberflächenwasser für die Trinkwassergewinnung.

Mulder, C.P.H., A. Jumpponen, P. Högberg, and K. Huss-Danell. 2002. How plant diversity and legumes affect nitrogen dynamics in experimental grassland communities. Oecologia 133:412–421.

Müller-Westermeier, G., and W. Riecke. 2004. Die Witterung in Deutschland 2003. p. 71–78. *In* Klimastatusbericht. Deutscher Wetterdienst, Offenbach.

Newman, E.I. 1995. Phosphorus inputs to terrestrial ecosytems. J. Ecol. 83:713–726.

Olde Venterink, H., N.M. Pieterse, J.D.M. Belgers, M.J. Wassen, and O.D. de Ruiter. 2002. N, P, and K budgets along nutrient availability and productivity gradients in wetlands. Ecol. Appl. 12:1010–1026.

Phoenix, G.K., R.E. Booth, J.R. Leake, D.J. Read, J.P. Grime, and J.A. Lee. 2003. Effects of enhanced nitrogen deposition and phosphorus limitation on nitrogen budgets of semi-natural grasslands. Global Change Biol. 9:1309–1321.

Rihm, B., and D. Kurz. 2001. Deposition and critical loads of nitrogen in Switzerland. Water Air Soil Pollut. 130:1223–1228.

Roscher, C., J. Schumacher, J. Baade, W. Wilcke, G. Gleixner, W.W. Weisser, B. Schmid, and E.-D. Schulze. 2004. The role of biodiversity for element cycling and trophic interactions: An experimental approach in a grassland community. Basic Appl. Ecol. 5:107–121.

Roscher, C., V.M. Temperton, M. Scherer-Lorenzen, M. Schmitz, J. Schumacher, B. Schmid, N. Buchmann, W.W. Weisser, and E.-D. Schulze. 2005. Overyielding in experimental grassland communities-irrespective of species pool or spatial scale. Ecol. Lett. 8:419–429.

Scherber, C., P.N. Mwangi, V.M. Temperton, C. Roscher, J. Schumacher, B. Schmid, and W.W. Weisser. 2006. Effects of plant diversity on invertebrate herbivory in experimental grassland. Oecologia 147: 489–500.

Scherer-Lorenzen, M., C. Palmborg, A. Prinz, and E.-D. Schulze. 2003. The role of plant diversity and composition for nitrate leaching in grasslands. Ecology 84:1539–1552.

Schulze, E.-D., O.L. Lange, and R. Oren. 1989. Forest decline and air pollution: A study of spruce (*Picea abies*) on acid soils. Springer, Berlin, Germany.

Schulze, E.-D., and H.A. Mooney. 1993. Biodiversity and ecosystem function. Springer, Berlin, Germany.

Simek, M., L. Jisova, and D.W. Hopkins. 2002. What is the so-called optimum pH for denitrification in soil? Soil Biol. Biochem. 34: 1227–1234.

Spehn, E.M., M. Scherer-Lorenzen, B. Schmid, A. Hector, M.C. Caldeira, P.G. Dimitrakopoulos, J.A. Finn, A. Jumpponen, G. O'Donnovan, J.S. Pereira, E.-D. Schulze, A.Y. Troumbis, and C. Körner. 2002. The role of legumes as a component of biodiversity in a cross-European study of grassland biomass nitrogen. Oikos 98:205–218.

SPSS. 2003. SPSS Version 11.5. SPSS, Chicago, IL.

Stadler, B., and B. Michalzik. 2000. Effects of phytophagous insects on micro-organisms and throughfall chemistry in forested ecosystems: Herbivores as switches for the nutrient dynamics in the canopy. Basic Appl. Ecol. 1:109–116.

Tilman, D., D. Wedin, and J. Knops. 1996. Productivity and sustainability influenced by biodiversity in grassland ecosystems. Nature 379:718–720.

Tilsner, J., N. Wrage, J. Lauf, and G. Gebauer. 2003. Emission of gaseous nitrogen oxides from an extensively managed grassland in NE Bavaria, Germany: I. Annual budgets of N_2O and NO_x emissions. Biogeochemistry 63:229–247.

TLUG. 2004. Fachkapitel Umweldaten 2004. *In* Fachkapitel Umweldaten, Jena, p. 1–93.

TLVwA. 2004. Thüringer Wassergesetz (ThüWG). TLVwA, Weimar, Germany.

Toor, G.S., L.M. Condron, H.J. Di, K.C. Cameron, and B.J. Cade-Menun. 2003. Characterization of organic phosphorus in leachate from a grassland soil. Soil Biol. Biochem. 35:1317–1323.

Ulrich, B. 1983. Interactions of forest canopies with atmospheric constituents: SO_2, alkali and earth alkali cations, and chloride. p. 33–45. *In* B. Ulrich and J. Pankrath ed. Effects of accumulation of air pollutants in forest ecosystems. D. Reidel Publishing, Dordrecht, The Netherlands.

Wassen, M.J., H. Olde Venterink, E.D. Lapshina, and F. Tanneberger. 2005. Endangered plants persist under phosphorus limitation. Nature 437:547–550.

Whitehead, D.C. 1995. Grassland nitrogen. CAB International, Wallingford.

Zak, D.R., K.S. Pregitzer, W.E. Holmes, A.J. Burton, and G.P. Zogg. 2004. Anthropogenic N deposition and the fate of (NO_3^-)-N-15 in a northern hardwood ecosystem. Biogeochemistry 69:143–157.

APPENDIX

Appendix 1: List of plant species, their abbreviations, and the according functional plant group used in the experiment.

Name of plant species	Abbreviation	Functional group	Name of plant species	Abbreviation	Functional group
Achillea millefolium agg. L.	Ach mil	tall herb	*Leucanthemum vulgare agg.* Lamk.	Leu vul	tall herb
Ajuga reptans L.	Aju rep	small herb	*Lotus corniculatus* L.	Lot cor	legume
Alopecurus pratensis L.	Alo pra	grass	*Luzula campestris* L. (Dc.)	Luz cam	grass
Anthoxanthum odoratum L.	Ant odo	grass	*Medicago lupulina* L.	Med lup	legume
Anthriscus sylvestris (L.) Hoffm.	Ant syl	tall herb	*Medicago X varia* Martyn	Med var	legume
Arrhenatherum elatius (L.) P. Beauv.	Arr ela	grass	*Onobrychis viciifolia* Scop.	Ono vic	legume
Avenula pubescens (Huds.) Holub.	Ave pub	grass	*Pastinaca sativa* L.	Pas sat	tall herb
Bellis perennis L.	Bel per	small herb	*Phleum pratense* L.	Phl pra	grass
Bromus erectus Huds.	Bro ere	grass	*Pimpinella major* (L.) Huds.	Pim maj	tall herb
Bromus hordeaceus L.	Bro hor	grass	*Plantago lanceolata* L.	Pla lan	small herb
Campanula patula L.	Cam pat	tall herb	*Plantago media* L.	Pla med	small herb
Carum carvi L.	Car car	tall herb	*Poa pratensis* L.	Poa pra	grass
Cardamine pratensis L.	Car pra	tall herb	*Poa trivialis* L.	Poa tri	grass
Centaurea jacea ssp. jacea L.	Cen jac	tall herb	*Primula veris* L.	Pri ver	small herb
Cirsium oleraceum (L.) Scop.	Cir ole	tall herb	*Prunella vulgaris* L.	Pru vul	small herb
Crepis biennis L.	Cre bie	tall herb	*Ranunculus acris* L.	Ran acr	tall herb
Cynosurus cristatus L.	Cyn cri	grass	*Ranunculus repens* L.	Ran rep	small herb
Dactylis glomerata L.	Dac glo	grass	*Rumex acetosa* L.	Rum ace	tall herb
Daucus carota L.	Dau car	tall herb	*Sanguisorba officinalis* L.	San off	tall herb
Festuca pratensis Huds.	Fes pra	grass	*Taraxacum officinale agg.* Wiggers	Tar off	small herb
Festuca rubra L.	Fes rub	grass	*Tragopogon pratensis* L.	Tra pra	tall herb
Galium mollugo agg. L.	Gal mol	tall herb	*Trifolium campestre* Schreb.	Tri cam	legume
Geranium pratense L.	Ger pra	tall herb	*Trifolium dubium* Sibth.	Tri dub	legume
Glechoma hederacea L.	Gle hed	small herb	*Trisetum flavescens* (L.) P. Beauv.	Tri fla	grass
Heracleum sphondylium L.	Her sph	tall herb	*Trifolium fragiferum* L.	Tri fra	legume
Holcus lanatus L.	Hol lan	grass	*Trifolium hybridum* L.	Tri hyb	legume
Knautia arvensis (L.) J.M. Coult s.str.	Kna arv	tall herb	*Trifolium pratense* L.	Tri pra	legume
Lathyrus pratensis L.	Lat pra	legume	*Trifolium repens* L.	Tri rep	legume
Leontodon autumnalis L.	Leo aut	small herb	*Veronica chamaedris* L.	Ver cha	small herb
Leontodon hispidus L.	Leo his	small herb	*Vicia cracca* L.	Vic cra	legume

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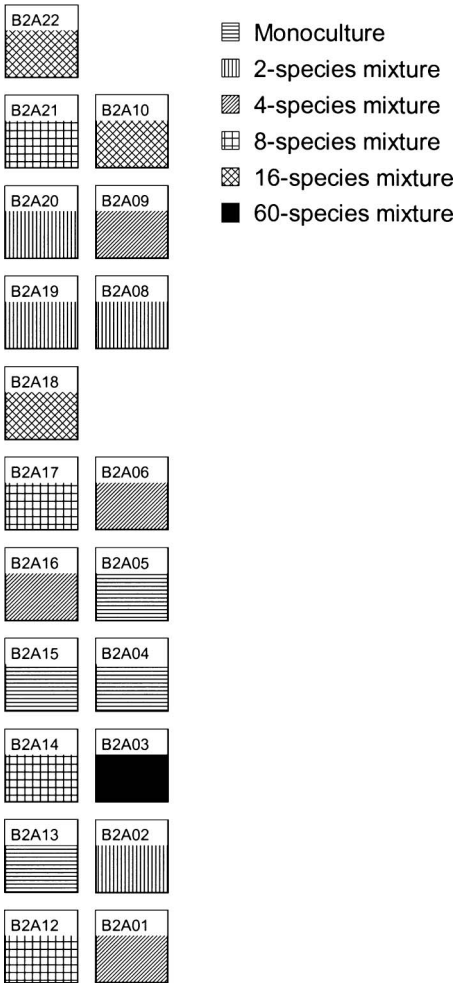

Appendix 2: Design of Block 2 of the experimental site.

Appendix 3: Plant species mixtures of the experimental plots. Abbreviations are explained in App. 1.

Plot name	Plant species in mixture
B2A01	Ant odo, Pru vul, Kna arv, Tri pra
B2A02	Fes rub, Tri fla
B2A03	Complete species pool
B2A04	Ger pra
B2A05	Fes pra
B2A06	Pla lan, Tar off, Lat pra, Med lup
B2A08	Ran acr, Tri cam
B2A09	Aju rep, Pla lan, Pri ver, Pru vul,
B2A10	Alo pra, Ant odo, Arr ela, Bro ere, Fes pra, Hol lan, Phl pra, Poa pra, Bel per, Leo aut, Leo his, Pla lan, Pri ver, Pru vul, Ran rep, Ver cha
B2A12	Ant syl, Gal mol, Ger pra, Her sph, Kna arv, Leu vul, Ran acr, San off
B2A13	Pla lan
B2A14	Luz cam, Phl pra, Leo his, Ver cha, Kna arv, San off, Tri dub, Tri hyb
B2A15	Ono vic
B2A16	Leo aut, Pla med, Kna arv, Vic cra
B2A17	Gle hed, Leo aut, Pla med, Tar off, Lat pra, Tri cam, Tri fra, Vic cra
B2A18	Alo pra, Bro hor, Cyn cri, Poa pra, Aju rep, Pla med, Pri ver, Ran rep, Ant syl, Cam pat, Car pra, Ger pra, Med lup, Tri cam, Tri dub, Tri rep
B2A19	Pla med, Tar off
B2A20	Pla lan, Tri dub
B2A21	Leo his, Pla med, Cre bie, Gal mol, San off, Lot cor, Med lup, Ono vic
B2A22	Cyn cri, Fes pra, Phl pra, Poa tri, Tri fla, Ach mil, Cam pat, Cen jac, Run ace, San off, Lat pra, Lot cor, Ono vic, Tri hyb, Tri rep, Vic cra

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